

12 Interrupts and events

This Section applies to the whole STM32F4xx family, unless otherwise specified.

12.1 Nested vectored interrupt controller (NVIC)

12.1.1 NVIC features

The nested vector interrupt controller NVIC includes the following features:

- 82 maskable interrupt channels for STM32F405xx/07xx and STM32F415xx/17xx, and up to 91 maskable interrupt channels for STM32F42xxx and STM32F43xxx (not including the 16 interrupt lines of Cortex®-M4 with FPU)
- 16 programmable priority levels (4 bits of interrupt priority are used)
- low-latency exception and interrupt handling
- power management control
- implementation of system control registers

The NVIC and the processor core interface are closely coupled, which enables low latency interrupt processing and efficient processing of late arriving interrupts.

All interrupts including the core exceptions are managed by the NVIC. For more information on exceptions and NVIC programming, refer to programming manual PM0214.

12.1.2 SysTick calibration value register

The SysTick calibration value is fixed to 18750, which gives a reference time base of 1 ms with the SysTick clock set to 18.75 MHz (HCLK/8, with HCLK set to 150 MHz).

12.1.3 Interrupt and exception vectors

See [Table 63](#) and [Table 62](#), for the vector table for the STM32F405xx/07xx and STM32F415xx/17xx and STM32F42xxx and STM32F43xxx devices.

12.2 External interrupt/event controller (EXTI)

The external interrupt/event controller consists of up to 23 edge detectors for generating event/interrupt requests. Each input line can be independently configured to select the type (interrupt or event) and the corresponding trigger event (rising or falling or both). Each line can also masked independently. A pending register maintains the status line of the interrupt requests.

The grey rows in the following tables describe the vectors without specific position.

Table 62. Vector table for STM32F405xx/07xx and STM32F415xx/17xx

Position	Priority	Type of priority	Acronym	Description	Address
-	-	-	-	Reserved	0x0000 0000
-	-3	fixed	Reset	Reset	0x0000 0004
-	-2	fixed	NMI	Non maskable interrupt. The RCC Clock Security System (CSS) is linked to the NMI vector.	0x0000 0008
-	-1	fixed	HardFault	All class of fault	0x0000 000C
-	0	settable	MemManage	Memory management	0x0000 0010
-	1	settable	BusFault	Pre-fetch fault, memory access fault	0x0000 0014
-	2	settable	UsageFault	Undefined instruction or illegal state	0x0000 0018
-	-	-	-	Reserved	0x0000 001C - 0x0000 002B
-	3	settable	SVCall	System service call via SWI instruction	0x0000 002C
-	4	settable	Debug Monitor	Debug Monitor	0x0000 0030
-	-	-	-	Reserved	0x0000 0034
-	5	settable	PendSV	Pendable request for system service	0x0000 0038
-	6	settable	SysTick	System tick timer	0x0000 003C
0	7	settable	WWDG	Window Watchdog interrupt	0x0000 0040
1	8	settable	PVD	PVD through EXTI line detection interrupt	0x0000 0044
2	9	settable	TAMP_STAMP	Tamper and TimeStamp interrupts through the EXTI line	0x0000 0048
3	10	settable	RTC_WKUP	RTC Wake-up interrupt through the EXTI line	0x0000 004C
4	11	settable	FLASH	Flash global interrupt	0x0000 0050
5	12	settable	RCC	RCC global interrupt	0x0000 0054
6	13	settable	EXTI0	EXTI Line0 interrupt	0x0000 0058
7	14	settable	EXTI1	EXTI Line1 interrupt	0x0000 005C
8	15	settable	EXTI2	EXTI Line2 interrupt	0x0000 0060
9	16	settable	EXTI3	EXTI Line3 interrupt	0x0000 0064
10	17	settable	EXTI4	EXTI Line4 interrupt	0x0000 0068
11	18	settable	DMA1_Stream0	DMA1 Stream0 global interrupt	0x0000 006C

Table 62. Vector table for STM32F405xx/07xx and STM32F415xx/17xx (continued)

Position	Priority	Type of priority	Acronym	Description	Address
12	19	settable	DMA1_Stream1	DMA1 Stream1 global interrupt	0x0000 0070
13	20	settable	DMA1_Stream2	DMA1 Stream2 global interrupt	0x0000 0074
14	21	settable	DMA1_Stream3	DMA1 Stream3 global interrupt	0x0000 0078
15	22	settable	DMA1_Stream4	DMA1 Stream4 global interrupt	0x0000 007C
16	23	settable	DMA1_Stream5	DMA1 Stream5 global interrupt	0x0000 0080
17	24	settable	DMA1_Stream6	DMA1 Stream6 global interrupt	0x0000 0084
18	25	settable	ADC	ADC1, ADC2 and ADC3 global interrupts	0x0000 0088
19	26	settable	CAN1_TX	CAN1 TX interrupts	0x0000 008C
20	27	settable	CAN1_RX0	CAN1 RX0 interrupts	0x0000 0090
21	28	settable	CAN1_RX1	CAN1 RX1 interrupt	0x0000 0094
22	29	settable	CAN1_SCE	CAN1 SCE interrupt	0x0000 0098
23	30	settable	EXTI9_5	EXTI Line[9:5] interrupts	0x0000 009C
24	31	settable	TIM1_BRK_TIM9	TIM1 Break interrupt and TIM9 global interrupt	0x0000 00A0
25	32	settable	TIM1_UP_TIM10	TIM1 Update interrupt and TIM10 global interrupt	0x0000 00A4
26	33	settable	TIM1_TRG_COM_TIM11	TIM1 Trigger and Commutation interrupts and TIM11 global interrupt	0x0000 00A8
27	34	settable	TIM1_CC	TIM1 Capture Compare interrupt	0x0000 00AC
28	35	settable	TIM2	TIM2 global interrupt	0x0000 00B0
29	36	settable	TIM3	TIM3 global interrupt	0x0000 00B4
30	37	settable	TIM4	TIM4 global interrupt	0x0000 00B8
31	38	settable	I2C1_EV	I ² C1 event interrupt	0x0000 00BC
32	39	settable	I2C1_ER	I ² C1 error interrupt	0x0000 00C0
33	40	settable	I2C2_EV	I ² C2 event interrupt	0x0000 00C4
34	41	settable	I2C2_ER	I ² C2 error interrupt	0x0000 00C8
35	42	settable	SPI1	SPI1 global interrupt	0x0000 00CC
36	43	settable	SPI2	SPI2 global interrupt	0x0000 00D0
37	44	settable	USART1	USART1 global interrupt	0x0000 00D4
38	45	settable	USART2	USART2 global interrupt	0x0000 00D8
39	46	settable	USART3	USART3 global interrupt	0x0000 00DC

Table 62. Vector table for STM32F405xx/07xx and STM32F415xx/17xx (continued)

Position	Priority	Type of priority	Acronym	Description	Address
40	47	settable	EXTI15_10	EXTI Line[15:10] interrupts	0x0000 00E0
41	48	settable	RTC_Alarm	RTC Alarms (A and B) through EXTI line interrupt	0x0000 00E4
42	49	settable	OTG_FS_WKUP	USB On-The-Go FS Wake-up through EXTI line interrupt	0x0000 00E8
43	50	settable	TIM8_BRK_TIM12	TIM8 Break interrupt and TIM12 global interrupt	0x0000 00EC
44	51	settable	TIM8_UP_TIM13	TIM8 Update interrupt and TIM13 global interrupt	0x0000 00F0
45	52	settable	TIM8_TRG_COM_TIM14	TIM8 Trigger and Commutation interrupts and TIM14 global interrupt	0x0000 00F4
46	53	settable	TIM8_CC	TIM8 Capture Compare interrupt	0x0000 00F8
47	54	settable	DMA1_Stream7	DMA1 Stream7 global interrupt	0x0000 00FC
48	55	settable	FSMC	FSMC global interrupt	0x0000 0100
49	56	settable	SDIO	SDIO global interrupt	0x0000 0104
50	57	settable	TIM5	TIM5 global interrupt	0x0000 0108
51	58	settable	SPI3	SPI3 global interrupt	0x0000 010C
52	59	settable	UART4	UART4 global interrupt	0x0000 0110
53	60	settable	UART5	UART5 global interrupt	0x0000 0114
54	61	settable	TIM6_DAC	TIM6 global interrupt, DAC1 and DAC2 underrun error interrupts	0x0000 0118
55	62	settable	TIM7	TIM7 global interrupt	0x0000 011C
56	63	settable	DMA2_Stream0	DMA2 Stream0 global interrupt	0x0000 0120
57	64	settable	DMA2_Stream1	DMA2 Stream1 global interrupt	0x0000 0124
58	65	settable	DMA2_Stream2	DMA2 Stream2 global interrupt	0x0000 0128
59	66	settable	DMA2_Stream3	DMA2 Stream3 global interrupt	0x0000 012C
60	67	settable	DMA2_Stream4	DMA2 Stream4 global interrupt	0x0000 0130
61	68	settable	ETH	Ethernet global interrupt	0x0000 0134
62	69	settable	ETH_WKUP	Ethernet Wake-up through EXTI line interrupt	0x0000 0138
63	70	settable	CAN2_TX	CAN2 TX interrupts	0x0000 013C
64	71	settable	CAN2_RX0	CAN2 RX0 interrupts	0x0000 0140

Table 62. Vector table for STM32F405xx/07xx and STM32F415xx/17xx (continued)

Position	Priority	Type of priority	Acronym	Description	Address
65	72	settable	CAN2_RX1	CAN2 RX1 interrupt	0x0000 0144
66	73	settable	CAN2_SCE	CAN2 SCE interrupt	0x0000 0148
67	74	settable	OTG_FS	USB On The Go FS global interrupt	0x0000 014C
68	75	settable	DMA2_Stream5	DMA2 Stream5 global interrupt	0x0000 0150
69	76	settable	DMA2_Stream6	DMA2 Stream6 global interrupt	0x0000 0154
70	77	settable	DMA2_Stream7	DMA2 Stream7 global interrupt	0x0000 0158
71	78	settable	USART6	USART6 global interrupt	0x0000 015C
72	79	settable	I2C3_EV	I ² C3 event interrupt	0x0000 0160
73	80	settable	I2C3_ER	I ² C3 error interrupt	0x0000 0164
74	81	settable	OTG_HS_EP1_OUT	USB On The Go HS End Point 1 Out global interrupt	0x0000 0168
75	82	settable	OTG_HS_EP1_IN	USB On The Go HS End Point 1 In global interrupt	0x0000 016C
76	83	settable	OTG_HS_WKUP	USB On The Go HS Wake-up through EXTI interrupt	0x0000 0170
77	84	settable	OTG_HS	USB On The Go HS global interrupt	0x0000 0174
78	85	settable	DCMI	DCMI global interrupt	0x0000 0178
79	86	settable	CRYP	CRYP crypto global interrupt	0x0000 017C
80	87	settable	HASH_RNG	Hash and Rng global interrupt	0x0000 0180
81	88	settable	FPU	FPU global interrupt	0x0000 0184

Table 63. Vector table for STM32F42xxx and STM32F43xxx

Position	Priority	Type of priority	Acronym	Description	Address
-	-	-	-	Reserved	0x0000 0000
-	-3	fixed	Reset	Reset	0x0000 0004
-	-2	fixed	NMI	Non maskable interrupt, Clock Security System	0x0000 0008

Table 63. Vector table for STM32F42xxx and STM32F43xxx (continued)

Position	Priority	Type of priority	Acronym	Description	Address
-	-1	fixed	HardFault	All class of fault	0x0000 000C
-	0	settable	MemManage	Memory management	0x0000 0010
-	1	settable	BusFault	Pre-fetch fault, memory access fault	0x0000 0014
-	2	settable	UsageFault	Undefined instruction or illegal state	0x0000 0018
-	-	-	-	Reserved	0x0000 001C - 0x0000 002B
-	3	settable	SVCall	System Service call via SWI instruction	0x0000 002C
-	4	settable	Debug Monitor	Debug Monitor	0x0000 0030
-	-	-	-	Reserved	0x0000 0034
-	5	settable	PendSV	Pendable request for system service	0x0000 0038
-	6	settable	Systick	System tick timer	0x0000 003C
0	7	settable	WWDG	Window Watchdog interrupt	0x0000 0040
1	8	settable	PVD	PVD through EXTI line detection interrupt	0x0000 0044
2	9	settable	TAMP_STAMP	Tamper and TimeStamp interrupts through the EXTI line	0x0000 0048
3	10	settable	RTC_WKUP	RTC Wake-up interrupt through the EXTI line	0x0000 004C
4	11	settable	FLASH	Flash global interrupt	0x0000 0050
5	12	settable	RCC	RCC global interrupt	0x0000 0054
6	13	settable	EXTI0	EXTI Line0 interrupt	0x0000 0058
7	14	settable	EXTI1	EXTI Line1 interrupt	0x0000 005C
8	15	settable	EXTI2	EXTI Line2 interrupt	0x0000 0060
9	16	settable	EXTI3	EXTI Line3 interrupt	0x0000 0064
10	17	settable	EXTI4	EXTI Line4 interrupt	0x0000 0068
11	18	settable	DMA1_Stream0	DMA1 Stream0 global interrupt	0x0000 006C
12	19	settable	DMA1_Stream1	DMA1 Stream1 global interrupt	0x0000 0070
13	20	settable	DMA1_Stream2	DMA1 Stream2 global interrupt	0x0000 0074
14	21	settable	DMA1_Stream3	DMA1 Stream3 global interrupt	0x0000 0078
15	22	settable	DMA1_Stream4	DMA1 Stream4 global interrupt	0x0000 007C
16	23	settable	DMA1_Stream5	DMA1 Stream5 global interrupt	0x0000 0080
17	24	settable	DMA1_Stream6	DMA1 Stream6 global interrupt	0x0000 0084

Table 63. Vector table for STM32F42xxx and STM32F43xxx (continued)

Position	Priority	Type of priority	Acronym	Description	Address
18	25	settable	ADC	ADC1, ADC2 and ADC3 global interrupts	0x0000 0088
19	26	settable	CAN1_TX	CAN1 TX interrupts	0x0000 008C
20	27	settable	CAN1_RX0	CAN1 RX0 interrupts	0x0000 0090
21	28	settable	CAN1_RX1	CAN1 RX1 interrupt	0x0000 0094
22	29	settable	CAN1_SCE	CAN1 SCE interrupt	0x0000 0098
23	30	settable	EXTI9_5	EXTI Line[9:5] interrupts	0x0000 009C
24	31	settable	TIM1_BRK_TIM9	TIM1 Break interrupt and TIM9 global interrupt	0x0000 00A0
25	32	settable	TIM1_UP_TIM10	TIM1 Update interrupt and TIM10 global interrupt	0x0000 00A4
26	33	settable	TIM1_TRG_COM_TIM11	TIM1 Trigger and Commutation interrupts and TIM11 global interrupt	0x0000 00A8
27	34	settable	TIM1_CC	TIM1 Capture Compare interrupt	0x0000 00AC
28	35	settable	TIM2	TIM2 global interrupt	0x0000 00B0
29	36	settable	TIM3	TIM3 global interrupt	0x0000 00B4
30	37	settable	TIM4	TIM4 global interrupt	0x0000 00B8
31	38	settable	I2C1_EV	I ² C1 event interrupt	0x0000 00BC
32	39	settable	I2C1_ER	I ² C1 error interrupt	0x0000 00C0
33	40	settable	I2C2_EV	I ² C2 event interrupt	0x0000 00C4
34	41	settable	I2C2_ER	I ² C2 error interrupt	0x0000 00C8
35	42	settable	SPI1	SPI1 global interrupt	0x0000 00CC
36	43	settable	SPI2	SPI2 global interrupt	0x0000 00D0
37	44	settable	USART1	USART1 global interrupt	0x0000 00D4
38	45	settable	USART2	USART2 global interrupt	0x0000 00D8
39	46	settable	USART3	USART3 global interrupt	0x0000 00DC
40	47	settable	EXTI15_10	EXTI Line[15:10] interrupts	0x0000 00E0
41	48	settable	RTC_Alarm	RTC Alarms (A and B) through EXTI line interrupt	0x0000 00E4
42	49	settable	OTG_FS_WKUP	USB On-The-Go FS Wake-up through EXTI line interrupt	0x0000 00E8
43	50	settable	TIM8_BRK_TIM12	TIM8 Break interrupt and TIM12 global interrupt	0x0000 00EC

Table 63. Vector table for STM32F42xxx and STM32F43xxx (continued)

Position	Priority	Type of priority	Acronym	Description	Address
44	51	settable	TIM8_UP_TIM13	TIM8 Update interrupt and TIM13 global interrupt	0x0000 00F0
45	52	settable	TIM8_TRG_COM_TIM14	TIM8 Trigger and Commutation interrupts and TIM14 global interrupt	0x0000 00F4
46	53	settable	TIM8_CC	TIM8 Capture Compare interrupt	0x0000 00F8
47	54	settable	DMA1_Stream7	DMA1 Stream7 global interrupt	0x0000 00FC
48	55	settable	FSMC	FSMC global interrupt	0x0000 0100
49	56	settable	SDIO	SDIO global interrupt	0x0000 0104
50	57	settable	TIM5	TIM5 global interrupt	0x0000 0108
51	58	settable	SPI3	SPI3 global interrupt	0x0000 010C
52	59	settable	UART4	UART4 global interrupt	0x0000 0110
53	60	settable	UART5	UART5 global interrupt	0x0000 0114
54	61	settable	TIM6_DAC	TIM6 global interrupt, DAC1 and DAC2 underrun error interrupts	0x0000 0118
55	62	settable	TIM7	TIM7 global interrupt	0x0000 011C
56	63	settable	DMA2_Stream0	DMA2 Stream0 global interrupt	0x0000 0120
57	64	settable	DMA2_Stream1	DMA2 Stream1 global interrupt	0x0000 0124
58	65	settable	DMA2_Stream2	DMA2 Stream2 global interrupt	0x0000 0128
59	66	settable	DMA2_Stream3	DMA2 Stream3 global interrupt	0x0000 012C
60	67	settable	DMA2_Stream4	DMA2 Stream4 global interrupt	0x0000 0130
61	68	settable	ETH	Ethernet global interrupt	0x0000 0134
62	69	settable	ETH_WKUP	Ethernet Wake-up through EXTI line interrupt	0x0000 0138
63	70	settable	CAN2_TX	CAN2 TX interrupts	0x0000 013C
64	71	settable	CAN2_RX0	CAN2 RX0 interrupts	0x0000 0140
65	72	settable	CAN2_RX1	CAN2 RX1 interrupt	0x0000 0144
66	73	settable	CAN2_SCE	CAN2 SCE interrupt	0x0000 0148
67	74	settable	OTG_FS	USB On The Go FS global interrupt	0x0000 014C
68	75	settable	DMA2_Stream5	DMA2 Stream5 global interrupt	0x0000 0150
69	76	settable	DMA2_Stream6	DMA2 Stream6 global interrupt	0x0000 0154
70	77	settable	DMA2_Stream7	DMA2 Stream7 global interrupt	0x0000 0158
71	78	settable	USART6	USART6 global interrupt	0x0000 015C

Table 63. Vector table for STM32F42xxx and STM32F43xxx (continued)

Position	Priority	Type of priority	Acronym	Description	Address
72	79	settable	I2C3_EV	I ² C3 event interrupt	0x0000 0160
73	80	settable	I2C3_ER	I ² C3 error interrupt	0x0000 0164
74	81	settable	OTG_HS_EP1_OUT	USB On The Go HS End Point 1 Out global interrupt	0x0000 0168
75	82	settable	OTG_HS_EP1_IN	USB On The Go HS End Point 1 In global interrupt	0x0000 016C
76	83	settable	OTG_HS_WKUP	USB On The Go HS Wake-up through EXTI interrupt	0x0000 0170
77	84	settable	OTG_HS	USB On The Go HS global interrupt	0x0000 0174
78	85	settable	DCMI	DCMI global interrupt	0x0000 0178
79	86	settable	CRYP	CRYP crypto global interrupt	0x0000 017C
80	87	settable	HASH_RNG	Hash and Rng global interrupt	0x0000 0180
81	88	Settable	FPU	FPU global interrupt	0x0000 0184
82	89	settable	UART7	UART 7 global interrupt	0x0000 0188
83	90	settable	UART8	UART 8 global interrupt	0x0000 018C
84	91	settable	SPI4	SPI 4 global interrupt	0x0000 0190
85	92	settable	SPI5	SPI 5 global interrupt	0x0000 0194
86	93	settable	SPI6	SPI 6 global interrupt	0x0000 0198
87	94	settable	SAI1	SAI1 global interrupt	0x0000 019C
88	95	settable	LCD-TFT	LTDC global interrupt	0x0000 01A0
89	96	settable	LCD-TFT	LTDC global Error interrupt	0x0000 01A4
90	97	settable	DMA2D	DMA2D global interrupt	0x0000 01A8

12.2.1 EXTI main features

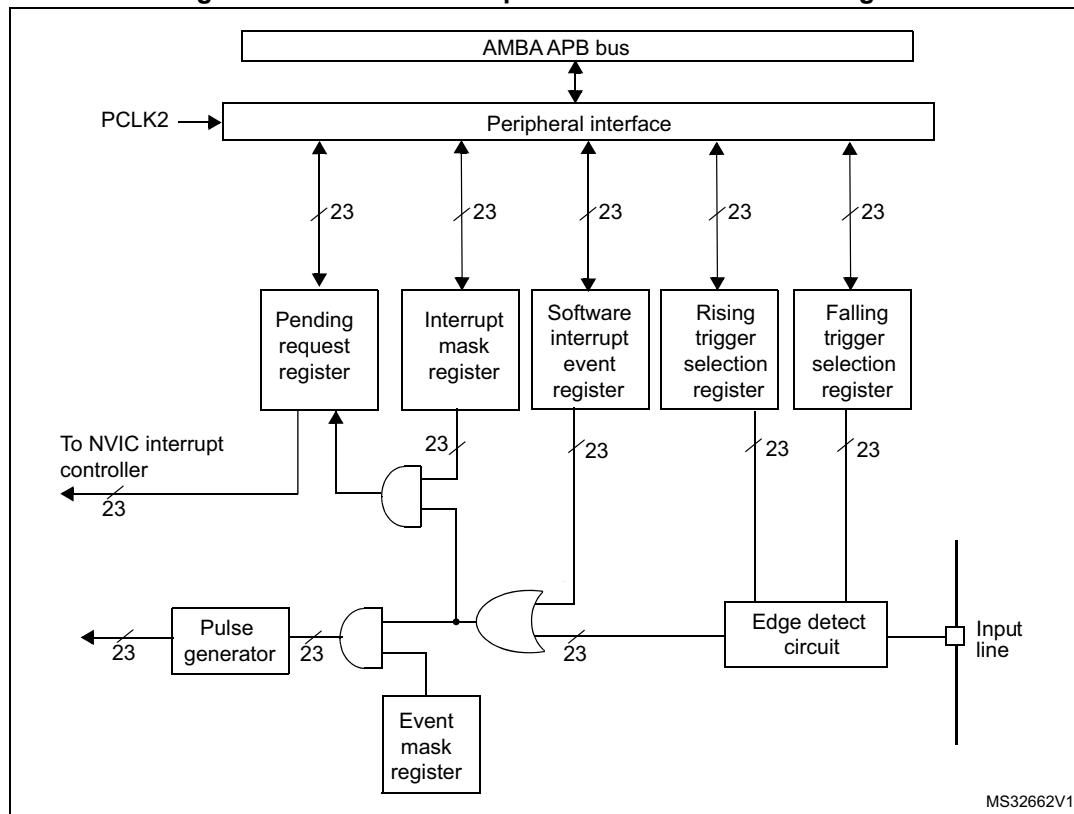
The main features of the EXTI controller are the following:

- independent trigger and mask on each interrupt/event line
- dedicated status bit for each interrupt line
- generation of up to 23 software event/interrupt requests
- detection of external signals with a pulse width lower than the APB2 clock period. Refer to the electrical characteristics section of the STM32F4xx datasheets for details on this parameter.

12.2.2 EXTI block diagram

Figure 41 shows the block diagram.

Figure 41. External interrupt/event controller block diagram



12.2.3 Wake-up event management

The STM32F4xx are able to handle external or internal events in order to wake up the core (WFE). The wake-up event can be generated either by:

- enabling an interrupt in the peripheral control register but not in the NVIC, and enabling the SEVONPEND bit in the Cortex®-M4 with FPU System Control register. When the MCU resumes from WFE, the peripheral interrupt pending bit and the peripheral NVIC IRQ channel pending bit (in the NVIC interrupt clear pending register) have to be cleared.
- or configuring an external or internal EXTI line in event mode. When the CPU resumes from WFE, it is not necessary to clear the peripheral interrupt pending bit or the NVIC IRQ channel pending bit as the pending bit corresponding to the event line is not set.

To use an external line as a wake-up event, refer to [Section 12.2.4: Functional description](#).

12.2.4 Functional description

To generate the interrupt, the interrupt line should be configured and enabled. This is done by programming the two trigger registers with the desired edge detection and by enabling the interrupt request by writing a '1' to the corresponding bit in the interrupt mask register. When the selected edge occurs on the external interrupt line, an interrupt request is

generated. The pending bit corresponding to the interrupt line is also set. This request is reset by writing a '1' in the pending register.

To generate the event, the event line should be configured and enabled. This is done by programming the two trigger registers with the desired edge detection and by enabling the event request by writing a '1' to the corresponding bit in the event mask register. When the selected edge occurs on the event line, an event pulse is generated. The pending bit corresponding to the event line is not set.

An interrupt/event request can also be generated by software by writing a '1' in the software interrupt/event register.

Hardware interrupt selection

To configure the 23 lines as interrupt sources, use the following procedure:

- Configure the mask bits of the 23 interrupt lines (EXTI_IMR)
- Configure the Trigger selection bits of the interrupt lines (EXTI_RTISR and EXTI_FTISR)
- Configure the enable and mask bits that control the NVIC IRQ channel mapped to the external interrupt controller (EXTI) so that an interrupt coming from one of the 23 lines can be correctly acknowledged.

Hardware event selection

To configure the 23 lines as event sources, use the following procedure:

- Configure the mask bits of the 23 event lines (EXTI_EMR)
- Configure the Trigger selection bits of the event lines (EXTI_RTISR and EXTI_FTISR)

Software interrupt/event selection

The 23 lines can be configured as software interrupt/event lines. The following is the procedure to generate a software interrupt.

- Configure the mask bits of the 23 interrupt/event lines (EXTI_IMR, EXTI_EMR)
- Set the required bit in the software interrupt register (EXTI_SWIER)

12.2.5 External interrupt/event line mapping

Up to 140 GPIOs (STM32F405xx/07xx and STM32F415xx/17xx), 168 GPIOs (STM32F42xxx and STM32F43xxx) are connected to the 16 external interrupt/event lines in the following manner:

Figure 42. External interrupt/event GPIO mapping((STM32F405xx/07xx and STM32F415xx/17xx)

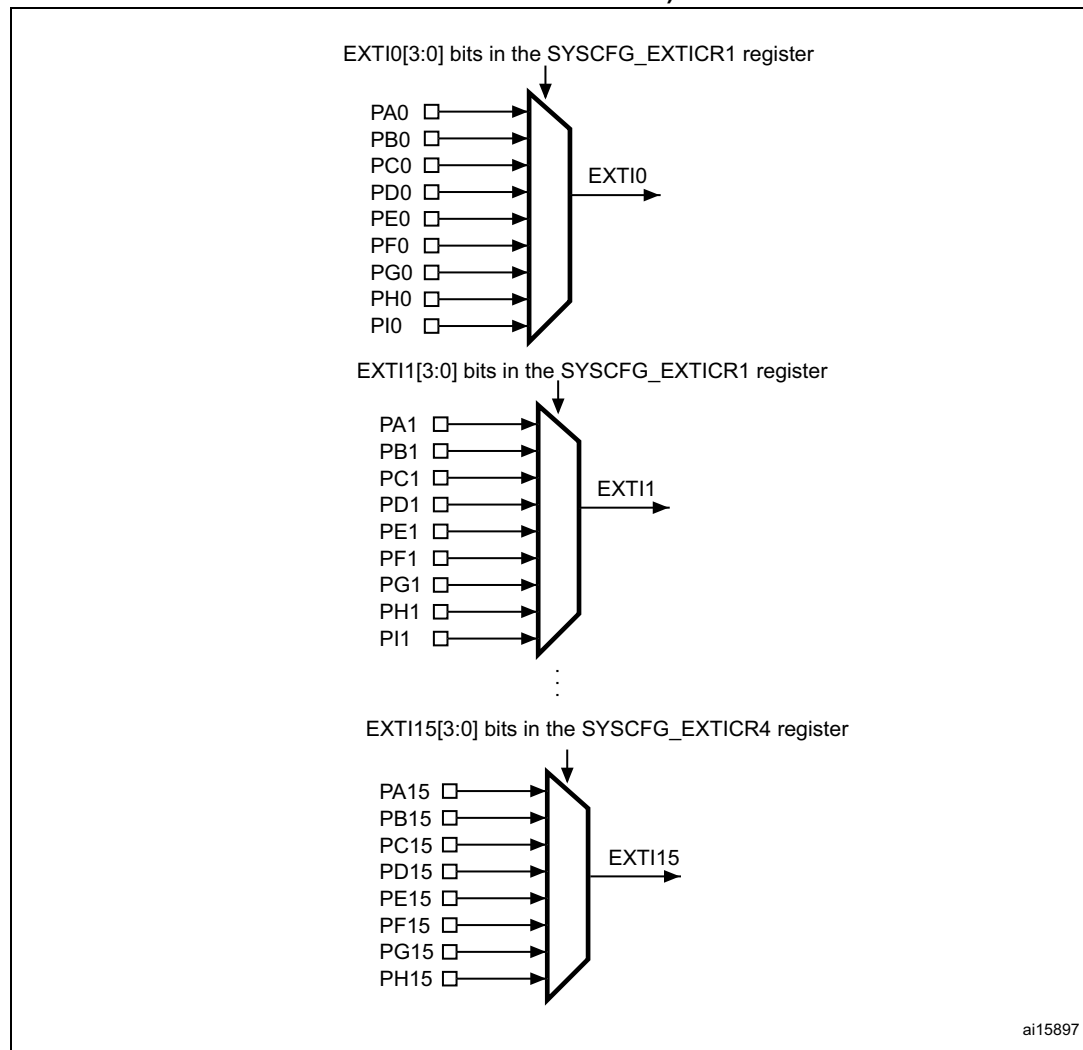
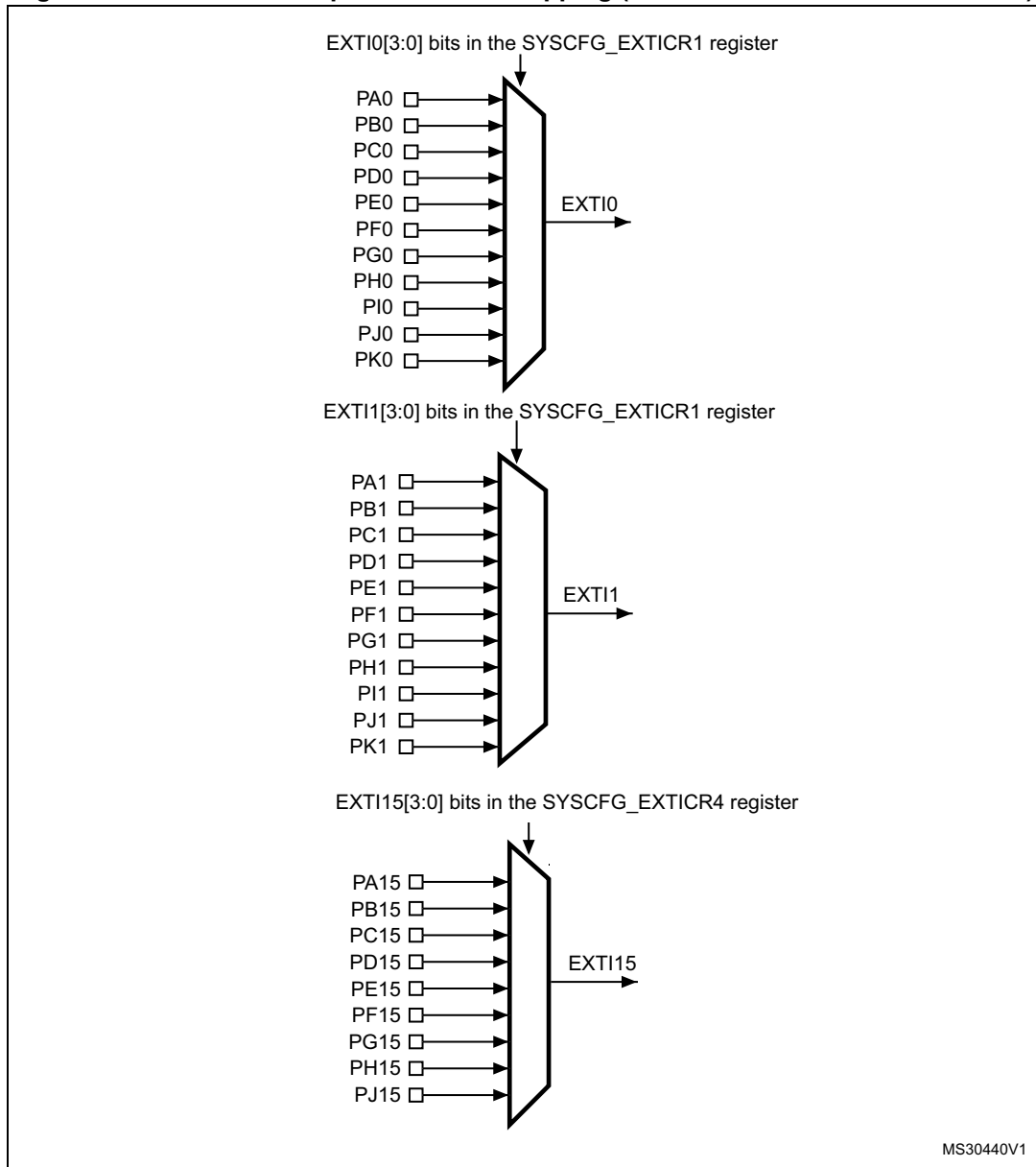


Figure 43. External interrupt/event GPIO mapping (STM32F42xxx and STM32F43xxx)

The seven other EXTI lines are connected as follows:

- EXTI line 16 is connected to the PVD output
- EXTI line 17 is connected to the RTC Alarm event
- EXTI line 18 is connected to the USB OTG FS Wake-up event
- EXTI line 19 is connected to the Ethernet Wake-up event
- EXTI line 20 is connected to the USB OTG HS (configured in FS) Wake-up event
- EXTI line 21 is connected to the RTC Tamper and TimeStamp events
- EXTI line 22 is connected to the RTC Wake-up event

12.3 EXTI registers

Refer to [Section 1.1: List of abbreviations for registers](#) for a list of abbreviations used in register descriptions.

12.3.1 Interrupt mask register (EXTI_IMR)

Address offset: 0x00

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved									MR22	MR21	MR20	MR19	MR18	MR17	MR16
									r/w	r/w	r/w	r/w	r/w	r/w	r/w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MR15	MR14	MR13	MR12	MR11	MR10	MR9	MR8	MR7	MR6	MR5	MR4	MR3	MR2	MR1	MR0
r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

Bits 31:23 Reserved, must be kept at reset value.

Bits 22:0 **MRx**: Interrupt mask on line x

0: Interrupt request from line x is masked

1: Interrupt request from line x is not masked

12.3.2 Event mask register (EXTI_EMR)

Address offset: 0x04

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved									MR22	MR21	MR20	MR19	MR18	MR17	MR16
									r/w	r/w	r/w	r/w	r/w	r/w	r/w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MR15	MR14	MR13	MR12	MR11	MR10	MR9	MR8	MR7	MR6	MR5	MR4	MR3	MR2	MR1	MR0
r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

Bits 31:23 Reserved, must be kept at reset value.

Bits 22:0 **MRx**: Event mask on line x

0: Event request from line x is masked

1: Event request from line x is not masked

12.3.3 Rising trigger selection register (EXTI_RTSR)

Address offset: 0x08

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved									TR22	TR21	TR20	TR19	TR18	TR17	TR16
									r/w	r/w	r/w	r/w	r/w	r/w	r/w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TR15	TR14	TR13	TR12	TR11	TR10	TR9	TR8	TR7	TR6	TR5	TR4	TR3	TR2	TR1	TR0
r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

Bits 31:23 Reserved, must be kept at reset value.

Bits 22:0 **TRx**: Rising trigger event configuration bit of line x

0: Rising trigger disabled (for Event and Interrupt) for input line

1: Rising trigger enabled (for Event and Interrupt) for input line

Note: *The external wake-up lines are edge triggered, no glitch must be generated on these lines. If a rising edge occurs on the external interrupt line while writing to the EXTI_RTISR register, the pending bit is be set.*

Rising and falling edge triggers can be set for the same interrupt line. In this configuration, both generate a trigger condition.

12.3.4 Falling trigger selection register (EXTI_FTSR)

Address offset: 0x0C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved									TR22	TR21	TR20	TR19	TR18	TR17	TR16
									r/w	r/w	r/w	r/w	r/w	r/w	r/w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TR15	TR14	TR13	TR12	TR11	TR10	TR9	TR8	TR7	TR6	TR5	TR4	TR3	TR2	TR1	TR0
r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

Bits 31:23 Reserved, must be kept at reset value.

Bits 22:0 **TRx**: Falling trigger event configuration bit of line x

0: Falling trigger disabled (for Event and Interrupt) for input line

1: Falling trigger enabled (for Event and Interrupt) for input line.

Note: *The external wake-up lines are edge triggered, no glitch must be generated on these lines. If a falling edge occurs on the external interrupt line while writing to the EXTI_FTSR register, the pending bit is not set.*

Rising and falling edge triggers can be set for the same interrupt line. In this configuration, both generate a trigger condition.

12.3.5 Software interrupt event register (EXTI_SWIER)

Address offset: 0x10

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved									SWIER 22	SWIER 21	SWIER 20	SWIER 19	SWIER 18	SWIER 17	SWIER 16
									rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SWIER 15	SWIER 14	SWIER 13	SWIER 12	SWIER 11	SWIER 10	SWIER 9	SWIER 8	SWIER 7	SWIER 6	SWIER 5	SWIER 4	SWIER 3	SWIER 2	SWIER 1	SWIER 0
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:23 Reserved, must be kept at reset value.

Bits 22:0 **SWIERx**: Software Interrupt on line x

If interrupt are enabled on line x in the EXTI_IMR register, writing '1' to SWIERx bit when it is set at '0' sets the corresponding pending bit in the EXTI_PR register, thus resulting in an interrupt request generation.

This bit is cleared by clearing the corresponding bit in EXTI_PR (by writing a 1 to the bit).

12.3.6 Pending register (EXTI_PR)

Address offset: 0x14

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved									PR22	PR21	PR20	PR19	PR18	PR17	PR16
									rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PR15	PR14	PR13	PR12	PR11	PR10	PR9	PR8	PR7	PR6	PR5	PR4	PR3	PR2	PR1	PR0
rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1

Bits 31:23 Reserved, must be kept at reset value.

Bits 22:0 **PRx**: Pending bit

0: No trigger request occurred

1: selected trigger request occurred

This bit is set when the selected edge event arrives on the external interrupt line.

This bit is cleared by programming it to '1'.

12.3.7 EXTI register map

[Table 64](#) gives the EXTI register map and the reset values.

Table 64. External interrupt/event controller register map and reset values

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x00	EXTI_IMR	Reserved										MR[22:0]																					
	Reset value											0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x04	EXTI_EMR	Reserved										MR[22:0]																					
	Reset value											0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x08	EXTI_RTSTR	Reserved										TR[22:0]																					
	Reset value											0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x0C	EXTI_FTSR	Reserved										TR[22:0]																					
	Reset value											0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x10	EXTI_SWIER	Reserved										SWIER[22:0]																					
	Reset value											0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x14	EXTI_PR	Reserved										PR[22:0]																					
	Reset value											0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Refer to [Section 2.3: Memory map](#) for the register boundary addresses.